$$\tan\theta = \frac{8 \times 1}{100} = \frac{8}{12.5} = 0.08$$

* For small θ minute angles, hypotenuse can be approximately equal to the base.

07/01/26

Units And Dimensions

* $q = n \times u$, where,

q = Physical quantity.
n = number
u = unit

$q = n_1 u_1 = n_2 u_2 = n_3 u_3$

$n u$ = constant.

$\Rightarrow n \propto \frac{1}{u}$

* Fundamental quantities in SI units :-

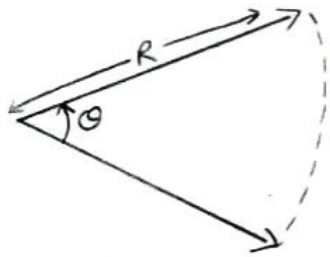
1,

	Fundamental quantity	SI unit	Dimension symbol
1,	Mass	kg	[M]
2,	Length	m	[L]
3,	Time	Sec	[T]
4,	Temperature	Kelvin	[Θ], [K]
5,	Current	Ampere	[A]
6,	Amount of Substance	Mole	[Mole]
7,	Luminous intensity	Candela	[Cd]

* Extra fundamental units that have been added:-

3.	2D Angle	Radians	$[M^0L^0T^0]$ (Dimensionless)
4.	3D Angle	Steradians	$[M^0L^0T^0]$ (Dimensionless)

* Plane Angle :-



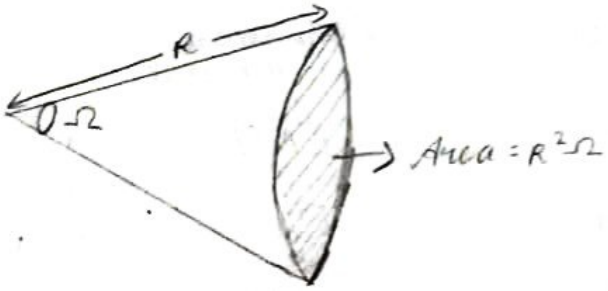
$s \longrightarrow$ Arc length

$$\theta = \frac{s}{R}$$

$$\theta_{max} = \frac{2\pi R}{R}$$

$$= 2\pi \text{ radians}$$

* Solid Angle / 3D Angle :-



$$\Omega = \frac{\text{Area}}{\text{Radius}^2}$$

$$\Omega_{max} = \frac{4\pi R^2}{R^2}$$

$$= 4\pi \text{ steradians}$$

* All other quantities derived from fundamental quantities are called derived quantities

Ex- Speed - m/s.
Density - kg/m^3

* **Dimensions of various quantities :-**

1, **Speed** = $\frac{\text{Distance}}{\text{Time}} = \frac{[L]}{[T]} = [LT^{-1}]$ (OR) $[M^0L^1T^{-1}]$.

2, **Velocity** = $\frac{\text{Displacement}}{\text{Time}} = \frac{[L]}{[T]} = [LT^{-1}]$

3, **Acceleration** = $\frac{m}{s^2} = \frac{[L]}{[T^2]} = [LT^{-2}]$

4, **Area** = $m^2 = [L^2]$

5, **Volume** = $m^3 = [L^3]$

6, **Density** = $kg/m^3 = \frac{[M]}{[L^3]} = [ML^{-3}]$

7, **Frequency** = $\frac{\text{no. of oscillations}}{\text{total time}} = \frac{1}{[T]} = [T^{-1}]$

8, **Force** = mass x acceleration = $[M] \times [LT^{-2}] = [MLT^{-2}]$

9, **Momentum** = mass x velocity = $[M] \times [LT^{-1}] = [MLT^{-1}]$

10, **Work done** = Force x displacement = $[MLT^{-2}] \times [L]$
$= [ML^2T^{-2}]$

11, **Kinetic energy** = $\frac{1}{2}mv^2 = \frac{1}{2}[M][LT^{-1}]^2 = [ML^2T^{-2}]$

12, **Torque** = Force x distance = $[MLT^{-2}] \times [L] = [ML^2T^{-2}]$

Note :- Work done, energy, and torque have the same dimensions.

13. **Pressure** = $\frac{\text{Force}}{\text{Area}} = \frac{MLT^{-2}}{L^2} = [ML^{-1}T^{-2}]$

14. **Stress** = $\frac{\text{Force}}{\text{Area}} = [ML^{-1}T^{-2}]$

15. **Power** = $\frac{\text{work done}}{\text{time}} = \frac{\text{energy}}{\text{time}} = \frac{[ML^2T^{-2}]}{T} = [ML^2T^{-3}]$

16. **Strain** = $\frac{\text{Change in length}}{\text{Original length}} = \frac{[L]}{[L]} = [M^0L^0T^0]$

17. **Refractive index** = $\frac{\text{Speed of light in vacuum}}{\text{Speed of light in given medium}}$

$= \frac{[LT^{-1}]}{[LT^{-1}]} = [M^0L^0T^0]$

18. **Relative density** = $\frac{\text{density of substance}}{\text{density of water}} = [M^0L^0T^0]$

19. **Universal gravitational constant (G)** = $\frac{\text{Force} \times R^2}{M_1 \times M_2}$

$= \frac{[MLT^{-2}] \times L^2}{M \times M} = [M^{-1}L^3T^{-2}]$

20. **acceleration due to gravity (g)** = $9.8\ m/s^2 = \frac{L}{T^2}$

$= [LT^{-2}]$

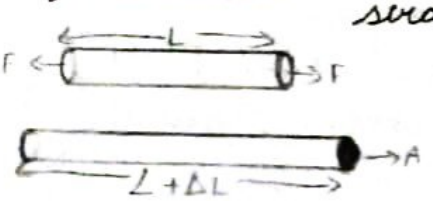
21. **Time Period** = $[T]$

22. **Wavelength** = distance from crest to crest = $[L]$

23. **Modulus of Elasticity** = $\frac{\text{Stress}}{\text{Strain}} = \frac{ML^{-1}T^{-2}}{M^0L^0T^0}$

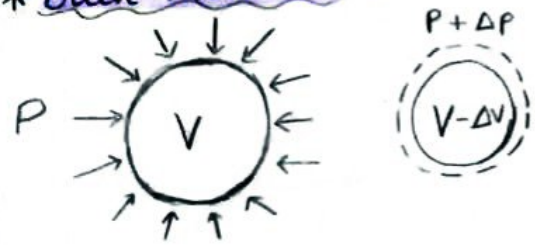
→ They are of different types.

* **Young's modulus** → $\frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L} = [ML^{-1}T^{-2}]$



ΔL = Change in length
L = Original length
A = Crosssectional area

* **Bulk modulus →**



ΔP = Increase in Pressure
ΔV = decrease in volume
V = Original volume

$$B_{modulus} = \frac{-\Delta P}{\Delta V/V} = [ML^{-1}T^{-2}]$$

* **Rigid modulus →** $[ML^{-1}T^{-2}]$

24, **Heat** $= [ML^2T^{-2}]$

25, **Heat capacity** $= \frac{\text{Heat absorbed}}{\text{increase in temperature}} = \frac{[ML^2T^{-2}]}{[K]}$

$= [ML^2T^{-2}K^{-1}]$

26, **Specific heat capacity** $= \frac{Q}{m\Delta\theta}$

$= \frac{\text{Heat absorbed}}{\text{mass} \times \text{change in tem}}$

$= \frac{ML^2T^{-2}}{MK} = [L^2T^{-2}K^{-1}]$

27, **Weight** $= \text{Force due to gravity}$
$W = mg$

08/01/26

Q) Find the dimensions of viscosity if its unit is pascal seconds.

= Pressure $\frac{N}{m^2} \times$ Sec

$= \frac{[MLT^{-2}][T]}{[L^2]}$

$= [ML^{-1}T^{-1}]$

Q) What are the dimensions of surface tension?

$= T = \frac{F}{\ell} = \frac{N}{M} = \frac{[MLT^{-2}]}{L}$

$= [MT^{-2}]$

8) what are the dimensions of stefan's constant?

$$P = \sigma A T^4$$

$$\sigma = \frac{P}{A T^4}$$

$$\sigma = \frac{[ML^2 T^{-3}]}{[L^2 K^4]}$$

$$= [MT^{-3} K^{-4}]$$

$$\therefore \sigma = [MT^{-3} K^{-4}].$$

---

* Intensity is defined as the energy carried per unit area per unit time.

⇒ Intensity of wave :-

$$I = \frac{E}{At}$$

$$= \frac{[ML^2 T^{-2}]}{[L^2 T]}$$

$$= [MT^{-3}]$$

---

* Principle of Homogenity :-

All terms in an expression, which are added or subtracted must have the same ~~dimension~~. dimension.

Ex: i) $L + \alpha$, ii) $F - \beta$ } where, α and β must have the same dimensions as L and F.

iii) $\frac{a}{t^2} = FV + \frac{\beta}{x^2}$

$$0 = FV + \frac{\beta}{x^2} - \frac{a}{t^2}$$

where, F is force,
V = speed,
t = time,
x = displacement

$$FV \equiv \frac{\beta}{x^2} \equiv \frac{a}{t^2}$$

$$\beta = FVx^2$$
$$\equiv [MLT^{-2}][LT^{-1}][L^2]$$
$$\equiv [ML^4T^{-3}]$$

$$a = FVt^2 = [MLT^{-2}][LT^{-1}][T^2]$$
$$= [ML^2T^{-1}]$$

(iv) $y = \sin\left(\frac{kx}{t}\right)$

x is in m
t is in seconds
find y, k.

$$\theta = \left(\frac{kx}{t}\right)$$

$\frac{kx}{t}$ is dimensionless
$= [M^0L^0T^0]$
(OR),

$$\frac{kx}{t} = 1$$
$$kx = t$$
$$k = \frac{t}{x}$$
$$= \frac{T}{L}$$
$$k = [TL^{-1}]$$

Sin θ is dimensionless
θ is also dimensionless
$y = \sin\theta$
$= [M^0L^0T^0]$

* All angles and trigonometric ratios are dimensionless

(v) $y = A \sin(ax + bt + c)$. Find dimensions of a, b, c, y, if x is m, t is sec, A is in meters.

$\theta = (ax + bt + c)$.

$m = L$

$t = T$

$A = L$

$y = L \sin(bL + bT + c)$

$y = L \sin\theta$ $(\sin\theta = [M^0 L^0 T^0])$

$\boxed{y = L}$

$ax = bt = c = \theta$

ax, bt, c are all dimensionless.

$ax = 1$

$a = \frac{1}{x} = \frac{1}{L}$

$\boxed{a = [L^{-1}]}$

$bt = 1$

$b = \frac{1}{t}$

$\boxed{b = [T^{-1}]}$

$\boxed{c = 1}$

---

* Dimensionless quantities :-

$a = \sin(\alpha)$

$b = \tan(\beta)$

$c = \cos(\gamma)$

$d = \log(e)$

a^b → Power is dimensionless

→ base is dimensionless

Q) $V = P_e^{\alpha x^2 - \beta v}$

find dimensions of α, β, γ, if P is power, x is distance, V is speed.

$\alpha x^2 - \beta v$ is an exponent power

$\alpha x^2 \equiv \beta v \equiv 1$

$\alpha = \frac{1}{x^2} = \frac{1}{(2)}$

$\beta = \frac{1}{V} = \frac{1}{LT} = [L^{-1}T]$

Q) $\left(P + \frac{a}{V^2}\right)(V-b) = RT$, where p is pressure, V is volume, T is temperature. Find dimensions of a, b, and R.

$\left(P + \frac{a}{V^2}\right)(V-b) = RT$

$P = ML^{-1}T^{-2}$

$V = L^3$

$T = K$

$\left(ML^{-1}T^{-2} + \frac{a}{(L^3)^2}\right)(L^3 - b) = R(K)$

$R = \frac{\text{Pressure} \times \text{Volume}}{\text{Temperature}}$

$= \frac{[ML^{-1}T^{-2}] \times [L^3]}{K}$

$= [ML^2T^{-2}]$

$a = [ML^5T^{-2}]$

$b = [L^3]$

* Applications of Dimensions :-

→ Finding dimensions of unknown quantities

→ checking dimensional correctness of an expression or formula or equation.

→ Inventing new formulas.

Q) Check the correctness of formula:

$$m = \frac{Fx T}{V}$$

LHS = M

$$RHS = \frac{[MLT^{-2}][T]}{[LT^{-1}]}$$

$$= \frac{ML\cancel{T^{1}}}{\cancel{LT^{-1}}}$$

$$= M.$$

Hence it is correct. (dimensionally)

Q) Find the correctness of the formula:

$$S = ut + at^2.$$

LHS = L

$$RHS = [LT^{-1} \times T] + [LT^{-2}] \times [T^2]$$

$$= L + L$$

All three terms have the same dimension.

Hence, it is dimensionally correct but error in pure numbers cannot be detected.

---

* Disadvantages of dimensional analysis :-

→ It cannot find errors in pure numbers in a formula.

⇒ Inventing new formulae :-

Q) L is length of pendulum, g is acceleration due to gravity, T is time taken for one oscillation. Deduce a relation between them.

→ Assume,

$$T \propto L^x g^y$$

$$[T] = L^x [LT^{-2}]^y$$

$$[T] = L^{x+y} T^{-2y}$$

$$L^0 T^1 = L^{x+y} T^{-2y}$$

$$x + y = 0$$

$$x = \frac{1}{2}$$

$$-2y = 1$$

$$y = -\frac{1}{2}$$

$$T \propto L^{1/2} g^{-1/2}$$

$$T \propto \sqrt{\frac{L}{g}}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

* 4th Application :-

→ It helps in conversion of units.

Q) Deduce a formula for the time period of a satellite given in terms of G and R, M.

where, G = universal gravitational constant

R = distance of satellite from the centre of the Earth

M = Mass of the Earth

→ Assume, the formula is:

$$T = G^x M^y r^z$$

$$[T] = [M^{-1} L^3 T^{-2}]^x [M]^y [L]^z$$

$$[T] = M^{-x} L^{3x} T^{-2x} M^y L^z$$

$$[M^0 L^0 T^1] = M^{-x+y} L^{3x+z} T^{-2x}$$

$$0 = -x + y$$

$$x = y$$

$$0 = 3x + z$$

$$\frac{-z}{3} = x$$

$$1 = -2x$$

$$x = \frac{1}{-2}$$

$$z = -3x$$

$$z = -3x - \frac{1}{2}$$

$$= \frac{3}{2}$$

$$y = -\frac{1}{2}$$

$$[T] = G^{-\frac{1}{2}} M^{-\frac{1}{2}} r^{\frac{3}{2}}$$

$$T = \frac{r^{3/2}}{\sqrt{GM}}$$

$$T = \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{r^3}{GM}}$$

→ Conversion of Units :-

9) Convert the following from MKS to CGS units

(i) 1J = ? ergs

$\hookrightarrow ML^2T^{-2}$

$= 1kg\ m^2\ Sec^{-2}$

$= 1000g\ (100\ cm)^2\ Sec^{-2}$

$= 10^7\ g\ cm^2\ Sec^{-2}$

$= 10^7$ ergs.

(ii) 1N = ? dynes ?

$\hookrightarrow MLT^{-2}$

$= 1kg\ m\ Sec^{-2}$

$= 1000g\ (100cm)\ Sec^{-2}$

$= 10^5\ Sec^{-2}$

$= 10^5$ dynes.

9) In a new system of units, 1 unit length = 10 m. 1 unit time = 5 secs. 1 unit mass = 2 kg. Convert newton into the new units.

Sol) $1N = MLT^{-2}$

$1N = 1kg/m/s^2$

$1kg' = 2kg$

$1m' = 10m$

$1Sec' = 5sec$

$1N = 1kg\ \frac{m}{Sec^2}$

$= \frac{\left(\frac{1}{2}kg'\right)\left(\frac{1}{10}m'\right)}{\left(\frac{1}{5}sec'\right)^2}$

$= \frac{\frac{1}{20}}{\frac{1}{25}}\ kg'm'/sec^2 = 1{\cdot}25\ N'$

Q) Find the value in new units of 20 J of energy, if 5 m = 2 unit length.

Sol) 1 kg = 100 unit mass

1 sec = 2 unit time

$$20\ J = \frac{ML^2T^{-2}}{}$$

$$1\ J = 1kg\ \frac{m^2}{sec^2}$$

$$20\ J = 20\ kg\ m^2/sec^2$$

$$20\ J = 20\ kg\ \frac{m^2}{s^2}$$

$$= \frac{\overset{10}{\cancel{20}} \times 100\ kg^1 \left(\frac{2}{5} m^1\right)^2}{(\cancel{2} sec^1)^2}$$

$$= \frac{\overset{40}{\cancel{200}}}{\cancel{1000}} \times \frac{4}{\cancel{25}}$$

$$= 80\ kg^1\ m^1/sec^2$$

Q) Convert $g = 9.8\ m/s^2$ to CGS units

$$g = 9.8\ \frac{m}{sec^2}$$

$$g = \frac{9.8 \times 100\ cm}{sec^2}$$

$$g = 980\ cm/s^2$$

Q) Density of water is $1 gm/cm^3$. Convert into SI unit.

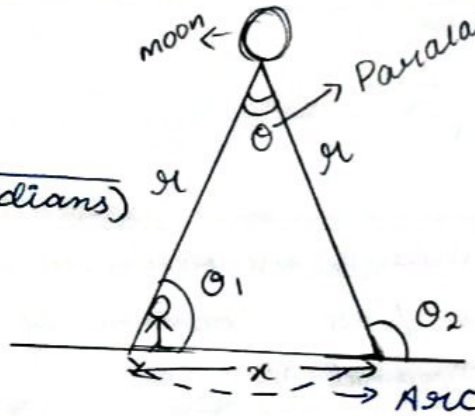
$$1g = \frac{1}{1000} kg = 10^{-3} kg$$

$1\ cm = \frac{1}{100}\ m = 10^{-2}\ m$

$\frac{1 \times 10^{-3}\ kg}{(10^{-2}\ m)^2} \rightarrow \frac{10^{-3}}{10^{-6}}$

$= 1000\ kg/m^2$

* **Parallax method:-**



$Arc = r\theta$

$r = \frac{Arc}{\theta (in\ radians)}$

$\theta = \theta_2 - \theta_1$

9) Angle subtended at the moon is 1° 54 minutes from 2 different points on earth which are diametrically opposite. If the diameter of earth is 1.276×10^7 m, find distance of the moon from earth

$\theta = 1^\circ 54'$

$r = ?$

Diameter $= 1.276 \times 10^7$ m

diameter $= r \times 10$

$\theta = 1^\circ + 54'$

$= 1^\circ + \left(\frac{54}{60}\right)^\circ$

$= (1.9)^\circ$

$1.276 \times 10^7 = r \times 1.9 \times \frac{22}{7} \times \frac{1}{180}$

$= r \times \frac{19}{10} \times \frac{22}{7} \times \frac{1}{1090}$

$$R = \frac{1.276 \times 10^7 \times 120 \times 45}{109 \times \frac{22}{7} \times 11}$$

$$= \frac{1.28 \times 10^7 \times 45 \times 7}{11}$$

$$= \frac{11.6}{128} \times 10^5 \times 45 \times 7$$

$$= 116 \times 10^4 \times 115 \times 7$$

$$= 4 \times 10^8.$$

9) If force, time, and velocity are taken to be fundamental quantities, find dimensions of energy in terms of F, T, and V.

Let Energy $= F^x T^y V^z$.

$$[ML^2T^{-2}] = [MLT^{-2}]^x [T]^y \left[\frac{L}{T}\right]^z$$

$$ML^2T^{-2} = M^x L^{x+z} T^{-2x+y-z}$$

$1 = x$

$2 = x + z$

$-2 = -2x + y - z$

$\boxed{x = 1}$

$2 = 1 + z$

$\boxed{z = 1}$

$-2 = -2 + y - 1$

$-2 = -3 + y$

$-2 + 3 = y$

$\boxed{1 = y}$

⇒ Find viscosity dimensions in terms of F, T, V.

⇒ If angular diameter of sun is 1920 arc seconds and distance of sun is 150 million km, find diameter of sun.

Ans: 1.4×10^9 approx

8) Find viscosity in terms of F, T, and V.

Let viscosity $= F^x T^y V^z$

$$[ML^{-1}T^{-1}] = M^x L^{x+z} T^{-2x+y-z}$$

$x = 0$

$x + z = -1$

$-2x + y - z = -1$

$\therefore \boxed{x = 0}$

$z = -1 - 0$

$z = -1$

$y = -1 + z + 2x$

$= -1 + (-1) + 2(0)$

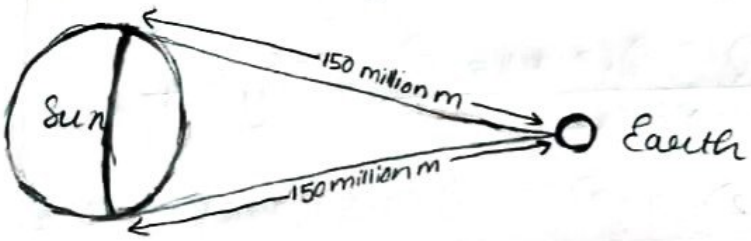
$= -1 - 1 + 0$

$= -2$

$\boxed{y = -2}$

---

9) If angular diameter of the sun 1920 arc seconds and distance of sun from earth is 150 million meters, find diameter of the sun.



$S = r\theta$

$d = 150 \times 10^6 \times 10^3 \times \theta$

$= 150 \times 10^9 \times \frac{1920}{3600 \times 180} \times \pi$

$= \frac{2 \times 2 \times 10^9 \times \pi}{9}$

$$= \frac{4 \times 10^9 \times 3.14}{9}$$

$$= \frac{1.4}{\cancel{12.56 \times 10^9}}{\cancel{9}}$$

$$= 1.4 \times 10^9 \text{ m}.$$

10/4/26

* Binomial expansion :-

$$(x+y)^n = x^n + nxy + \frac{n(n-1)}{1\times2} x^{n-2}y^2 + \frac{n(n-1)(n-2)}{1\times2\times3} \times x^{n-3}y^3 + \frac{n(n-1)(n-2)(n-3)}{1\times2\times3\times4} x^{n-4}y^4 \ldots + y^n$$

Ex: $(x+y)^7 = x^7 + 7xy + \frac{7(7-1)}{2} x^{7-2}y^2 + \frac{7(7-1)(7-2)}{2\times3} \times x^{7-3}y^3 + \frac{7(7-1)(7-2)(7-3)}{2\times3\times4} x^{7-4}y^4 + \frac{7(7-1)(7-2)(7-3)(7-4)}{2\times3\times4\times5} x^{7-5}y^5 + \frac{7(7-1)(7-2)(7-3)(7-4)(7-5)}{2\times3\times4\times5\times6} x^{7-6}y^6 + \frac{7(7-1)(7-2)(7-3)(7-4)(7-5)(7-6)}{2\times3\times4\times5\times6\times7} x^{7-7}y^7$

$$= x^7 + 7xy + \frac{(7)(6)}{2} x^5y^2 + \frac{7(6)(5)}{2\times3} x^4y^3 + \frac{7(6)(5)(4)}{2\times3\times4} x^3y^4 + \frac{(7)(6)(5)(4)(3)}{2\times3\times4\times5} x^2y^5 + \frac{7(6)(5)(4)(3)(2)}{2\times3\times4\times5\times6} x^1y^6 + \frac{7(6)(5)(4)(3)(2)(1)}{2\times3\times4\times5\times6\times7} x^0y^7$$

$= x^7 + 7xy + 21x^5y^2 + 35x^4y^3 + 35x^3y^4 + 21x^2y^5 + 7xy^6 + x^0y^7$

$= x^7 + 7xy + 21x^5y^2 + 35x^4y^3 + 35x^3y^4 + 21x^2y^5 + 7xy^6 + y^7$

* Binomial Pyramid :-

Q) Expand $(x+y)^4$

$= x^4 + 4xy + \frac{4(4-1)}{1\times 2} x^{4-2}y^2 + \frac{4(4-1)(4-2)}{1\times 2\times 3} \times x^{4-3}y^3 + \frac{4(4-1)(4-2)(4-3)}{1\times 2\times 3\times 4} x^{4-4}y^4$

$= x^4 + 4xy + \frac{4\times 3}{2} x^2y^2 + \frac{4(3)(2)}{6} x^1y^3 + \frac{4(3)(2)(1)}{(24)} x^0y^4$

$= x^4 + 4xy + 6x^2y^2 + 4xy^3 + y^4$

* Binomial pyramid :-

```
          0 1 0
        0 1 1 0
      0 1 2 1 0
    0 1 3 3 1 0
  0 1 4 6 4 1 0
0 1 5 10 10 5 1 0
0 1 6 15 20 15 6 1 0
0 1 7 21 35 35 21 7 1 0
0 1 8 28 56 70 56 28 8 1 0
0 1 9 36 84 126 126 84 36 9 1 0
```

Ex: $(1+x)^6$

$= 1^6 + 6x + \frac{6(5)}{2} x^4y^2 + \cdots$

$= 1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + x^6$

Ex: $(1+x)^5$

$= 1 + 5x + 10x^2 + 10x^3 + 5x^4 + 6x + x^5$

* Binomial approximation :-

If $x << 1$,

$(1+x)^n \cong 1+nx$.

Q) Find $\sqrt{0.99}$

$\sqrt{0.99} = \sqrt{1-0.01}$

$= (1-0.01)^{1/2}$

$= \left(1 - \frac{1}{2} \times 0.01\right)$

$= \left(\frac{2-1}{2} \times 0.01\right)$

$= \left(\frac{1}{2} \times \overset{0.995}{\cancel{0.01}}\right)$

$= 0.995$

(ii) $(1.01)^{365}$

$(1.01)^{365} = (1+0.01)^{365}$

$= (1 + 3.65)$

$= 4.65$.

(iii) $\sqrt[3]{0.97}$

$= (1-0.03)^{\frac{1}{3}}$

$= \left(1 - \frac{1}{3} \times \overset{0.01}{\cancel{0.03}}\right)$

$= 0.99$

(iv) $\sqrt{101}$

$= \sqrt{100(1+0.01)}$

Q) $\sqrt{\frac{1-x}{1+x}}$ where $x << 1$

$= \frac{\sqrt{1-x}}{\sqrt{1+x}}$

$= \sqrt{(1-x)(1+x)^{-1}}$

$= \sqrt{(1-x)(1-x)}$

$= \sqrt{(1-x)^2}$

$= (1-x)$

Q) $\frac{1}{r-a} - \frac{1}{r+a}$

$\frac{1}{r\left(1-\frac{a}{r}\right)} - \frac{1}{r\left(1+\frac{a}{r}\right)}$

$= \frac{1}{r}\left[\left(1-\frac{a}{r}\right)^{-1} - \left(\frac{a}{r}+1\right)^{-1}\right]$

$= \frac{1}{r}\left[\left(1+\frac{a}{r}\right) - \left(1-\frac{a}{r}\right)\right]$

$= \frac{1}{r}\left[\frac{2a}{r}\right]$

$= \frac{2a}{r^2}$

$= 10(1+0.01)^{1/2}$

$= 10\left(1+\frac{1}{2}\times 0.01\right)$

$= 10 + 0.05$

$= 10.05$

(v) $\sqrt[3]{1006}$

$= \sqrt[3]{1000+6}$

$= \sqrt[3]{1000\left(1+\frac{6}{1000}\right)}$

$= 10\left(1+\frac{6}{1000}\right)^{\frac{1}{3}}$

$= 10\left(1+\frac{2}{500}\right)$

$= 10 + 0.02$

$= 10.02$

(vi) $\sqrt[3]{1012}$

$= \sqrt[3]{1000+12}$

$= \sqrt[3]{1000+\left(1+\frac{12}{1000}\right)}$

$= 10\left(1+\frac{12}{1000}\right)^{\frac{1}{3}}$ (0.04)

$= 10.04.$

Q) find $\frac{x+a}{x-a}$

$= \frac{x(1+a/x)}{x(1-a/x)}$

if x is very large as compared to a.

$x >> a$

$\frac{a}{x} << \frac{x}{x}$

$\frac{a}{x} << 1$

$= (1+\frac{a}{x})(1-\frac{a}{x})^{-1}$

$= (1+\frac{a}{x})\left((1-(-1)\frac{a}{x}\right)$

$= \left(\frac{a}{x}+1\right)\left(1+\frac{a}{x}\right)$

$= \left(1+\frac{a}{x}\right)^2$

$= \left(1+2\frac{a}{x}\right)$

Q) $\sqrt{x^2+a^2}$

$= \sqrt{a^2\left(\frac{x^2}{a^2}+1\right)}$

$= a\left(1+\frac{x^2}{a^2}\right)^{1/2}$

$= a\left(1+\frac{1}{2}\frac{x^2}{a^2}\right)$

$= a\left(1+\frac{1}{2}\left(\frac{x}{a}\right)^2\right)$

$= a + \frac{x^2}{2a}$

* Arithemetic Progressions :-

n^{th} term $= a+(n-1)d$.

$d = T_2 - T_1$

$d = T_3 - T_2$

$S_n = \frac{n}{2}(2a+(n-1)d)$

$S_n = n \times \frac{(\text{first term} + \text{last term})}{2}$

Ex: find sum of first 12 terms.

$d = -1$

$a = 100$

$n = 12$

$S_n = 6(200 + 11 \times -1)$

$= 6(200 - 11)$

$= 6(189)$

$= 1134$

$= 40^{th}$ term

$= 100 + (40-1)(-1)$

$= 100 + (39)(-1)$

$= 100 - 39$

$= 61$

2) $S_n = 60 + \cdots + 124$

find S_n where $n = 12$

$S_n = n\left(\frac{a+an}{2}\right)$

$= 12 \times \frac{60+24}{2}$

$= 6 \times 84$

$= 504$

$504 = 6(120 + 11 \times d)$

$\frac{504}{6} = 84 = (120 + 11d)$

$84 - 120 = 11d$

$d = -\frac{36}{11}$

* Geometric Progressions :-

$a, ar, ar^2, ar^3, ar^4, \ldots ar^{n-1}$

$r = \frac{\text{second term}}{\text{first term}} = \frac{\text{third}}{\text{second}} \ldots \frac{T_{n+1}}{T_n}$.

→ Common ratio

$\Rightarrow S_n = a + ar + ar^2 \ldots ar^{n-1}$

$\Rightarrow S_n = a(1 + r + r^2 \ldots r^{n-1})$

$\Rightarrow S_n = a\left(\frac{1-r^n}{1-r}\right)$

$\Rightarrow \frac{x^n - 1}{x-1} = x^{n-1} + x^{n-2} + \ldots 4x^2 + x + 1$

$\Rightarrow \frac{1-x^n}{1-x} = \frac{x^n - 1}{x-1}$

---

* **Sum of an infinite number of a GP:-**

(1) $r \geq 1$,

$S_n = a + ar + ar^2 \ldots \infty$

$S_n = 1 + 2 + 2^2 + 2^3 \ldots \infty$

$= \infty$

(2) $|r| < 1$,

$S_n = a + ar + ar^2 \ldots \infty$

$= \frac{a}{1-r}$.

Ex: $S_n = 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \ldots$

$r = \frac{1/2}{1} = 1/2$

$S_n = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = \frac{1}{\frac{1}{2}} = \frac{2}{1} = 2$.

Differentiation :-

$\Delta \rightarrow$ Delta

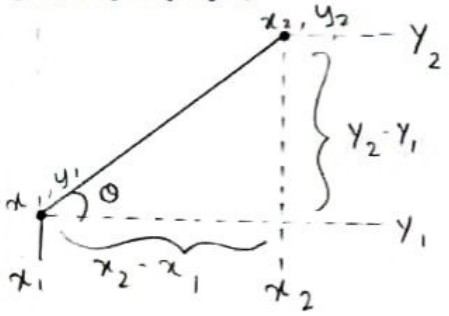
$\delta \rightarrow$ Del.

$\Delta x = x_{final} - x_{initial}$

= change in value of x.

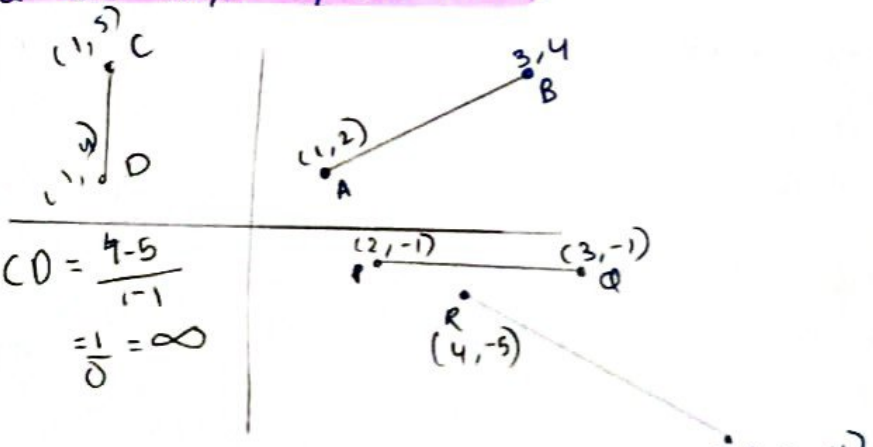
dx means infinitely small change in value of x.

Slope of a line :-



$$\text{Slope} = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

Q) Find the slope of the line :-



$CD = \frac{4-5}{1-1}$

$= \frac{1}{0} = \infty$

$AB = \frac{4-2}{3-1} = \frac{2}{2} = 1$

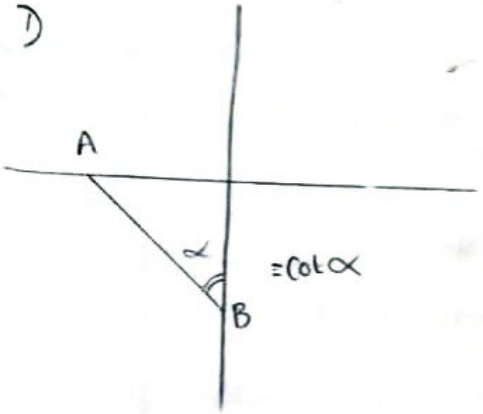
$PQ = \frac{-1+1}{3-2} = \frac{0}{1} = 0$

$RS = \frac{-11+5}{10-4} = \frac{-6}{6} = -1$

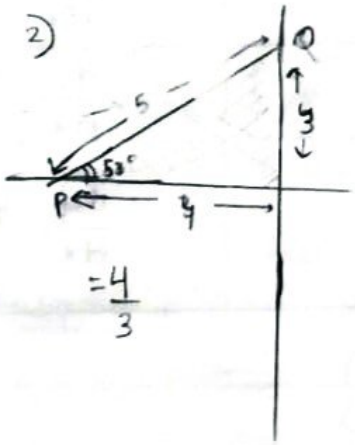
The slope of a vertical line is ∞ and that of a horizontal line is 0.

Q) Find the slope of the following

1)

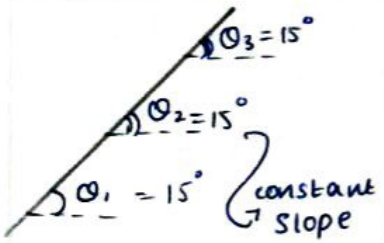


2)



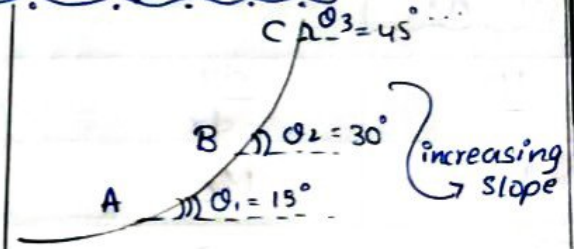
$= \frac{4}{3}$

Slope of a line vs Slope of a curve:-



Slope of a line at all points are constant.

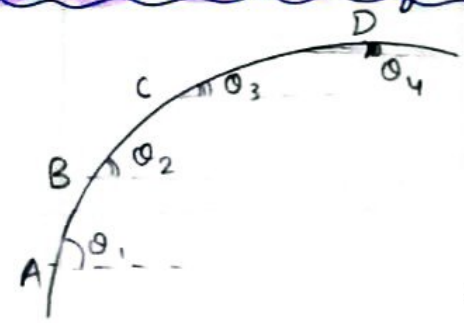
Slope is given by $\frac{\Delta y}{\Delta x}$



On a curved slope, θ is different at different points.

Slope is given by $\frac{dy}{dx}$

⇒ Curve with decreasing slope:-



*	y	$\frac{dy}{dx}$
1)	If y = constant	$\frac{dy}{dx} = 0$
2)	If $y = x$	$\frac{dy}{dx} = 1$
3)	If $y = x^2$	$\frac{dy}{dx} = 2x$

** ⇒

$x^2 = 2x$
$x^3 = 3x^2$
$x^4 = 4x^3$
$x^5 = 5x^4$
$x^n = nx^{n-1}$

S.No	y	$\frac{dy}{dx}$
1)	$\ln x$	$1/x$
2)	$\sin x$	$\cos x$
3)	$\cos x$	$-\sin x$
4)	$\tan x$	$\sec^2 x$
5)	$\sec x$	$\sec x \tan x$
6)	$\cot x$	$-\operatorname{cosec}^2 x$
7)	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
8)	e^x	e^x
9)	a^x	$a^x \ln a$

g) $y = x + x^2 + x^3 + x^5$

$\frac{dy}{dx} = 1 + 2x + 3x^2 + 5x^4$

Q) $y = \sin x + \ln x + \cos x$

$$\frac{dy}{dx} = 1/x + \cos x + (-\sin x)$$

$$= \frac{1}{x} + \cos x - \sin x$$

Q) $y = \sec x + \cot x + e^x$

$$\frac{dy}{dx} = \sec x \tan x + (-\operatorname{cosec}^2 x) + e^x$$

$$= \sec x \tan x - \operatorname{cosec}^2 x + e^x$$

Q) $y = \sqrt{x} + \frac{1}{\sqrt{x}}$

$$\frac{dy}{dx} = \sqrt{1} + \frac{1}{\sqrt{1}}$$

$$= 1^{1/2} + \frac{1}{1^{1/2}}$$

$$= 1^{1/2} + 1^{-1/2}$$

* Rules of differenciation:-

(i) $\frac{d}{dx}(3\sin x) = 3\frac{d}{dx}\sin x$

$$= 3\cos x$$

$$\frac{d}{dx}(4x^3) = 4\frac{d}{dx}x^3$$

$$= 4 \times 3x^2$$

$$= 12x^2$$

Example:- $\frac{d}{dx}[6x^5 + 7x^2 + \sqrt{2}\sqrt{x}]$

$= 6(5x^4) - 7(2x) + \sqrt{2}\left(\frac{1}{2\sqrt{x}}\right)$

$= 30x^4 - 14x + \frac{\sqrt{2}}{2\sqrt{x}}$

$\Rightarrow \frac{dy}{dx}$ is also called y or derivative of x.

Q) (i) $y = \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$

$y = x^{-1} + x^{-2} + x^{-3}$

$\frac{dy}{dx} = -1x^{-1-1} + -2x^{-2-1} - 3x^{-3-1}$

$= -x^{-2} - 2x^{-3} - 3x^{-4}$

$= \frac{-1}{x^2} - \frac{2}{x^3} - \frac{3}{x^4}$

(ii) $y = 4\sin x + 3\cos x$.

$\frac{dy}{dx} = 4\cos x + 3(-\sin x)$

$= 4\cos x - 3\sin x$

(iii) $y = \pi \ln x$

$\frac{dy}{dx} = \pi\left(\frac{1}{x}\right)$

$= \frac{\pi}{x}$

Q) $y = ax^4 + bx^3 + cx^2 + d$

find derivative of y

$$y' = \frac{dy}{dx} = a(4x^3) + b(3x^2) + c(2x)$$

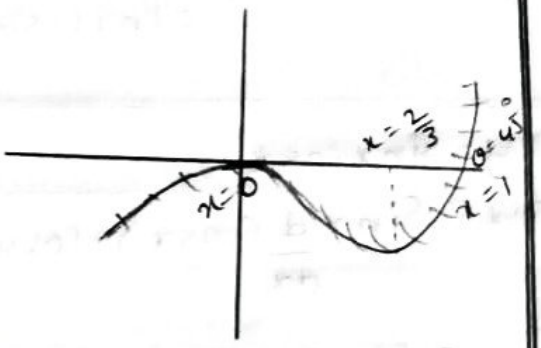
Q) If $y = x^3 - x^2$, find values of x for which the derivative of $y = 0$.

$$\frac{dy}{dx} = 3x^2 - 2x$$

$$\Rightarrow 3x^2 - 2x = 0$$

$$x(3x - 2) = 0$$

$$x = 0, x = \frac{2}{3}$$



Slope at $x = 1$ is $3 \times 1^2 = 2 \times 1$

$$= 3 - 2$$

$$= 1.$$

$$\tan\theta = \frac{dy}{dx} = 1$$

at $x = 1$

$\theta = 45°$.

Q) For what value of x is derivative of $\tan x = 1$?

$$\tan x \rightarrow \sec^2 x = 1$$

$$\frac{1}{\cos^2 x} = 1$$

$$\cos^2 x = 1$$

$$\cos x = \pm 1$$

at $x = 0$, $\cos x = 1$

$x = \pi$, $\cos\pi = -1$.

* Product Rule :-

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + u\frac{du}{dx}$$

⇒ **Example:** $\frac{d}{dx}(\ln x \cos x) = \ln x \frac{d}{dx}\cos x + \cos x \frac{d}{dx}\ln x$

$$= \ln x(-\sin x) + \cos x\left(\frac{1}{x}\right)$$

=

⇒ Q) (i) $y = \sin x \cos x$

$$= \frac{d}{dx}\sin x \cos x = \sin x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(\sin x)$$

$$= \sin x(-\sin x) + \cos x(\cos x)$$

$$= -\sin^2 x + \cos^2 x$$

$$= \cos 2x$$

$$\boxed{\cos^2 x - \sin^2 x = \cos 2x}.$$

(ii) $y = x \sin x$

$$\frac{dy}{dx} = x\frac{d}{dx}\sin x + \sin x \frac{dx}{dx}$$

$$= x(\cos x) + \sin x$$

(iii) $y = x \ln x$

$$\frac{dy}{dx} = x\frac{d}{dx}\ln x + \ln x \frac{dx}{dx}$$

$$= x \times \frac{1}{x} + \ln x \times 1$$

$$= 1 + \ln x$$

9) $y = (x)(\sin x \cos x)$

$$\frac{dy}{dx} = \frac{x\,d}{dx}\sin x\cos x + \sin x\cos x \frac{dx}{dx}$$

$$= x\left(\sin x \frac{d}{dx}\cos x + \cos x \frac{d}{dx}\sin x\right) + \sin x\cos x$$

$$= x(\sin x(-\sin x) + \cos x(\cos x)) + \sin x\cos x$$

$$= x(-\sin^2 x + \cos^2 x) + \sin x\cos x$$

$$= x(\cos 2x) + \sin x\cos x.$$

* Division Rule:-

$$\frac{d}{dx}\left(\frac{N}{D}\right) = \frac{D\frac{d}{dx}N - N\frac{d}{dx}D}{D^2}$$

Example:- $\frac{d}{dx}\frac{\sin x}{x} = \frac{x\frac{d}{dx}\sin x - \sin x\frac{d}{dx}x}{x^2}$

2) $y = \frac{x+2}{x+1}$

$$\frac{dy}{dx} = \frac{(x+1)\frac{d}{dx}(x+2) - (x+2)\frac{d}{dx}(x+1)}{(x+1)^2}$$

$$= \frac{(x+1)(1+0) - (x+2)(1+0)}{(x+1)^2}$$

$$= \frac{(x+1)-(x+2)}{(x+1)^2}$$

$$= \frac{-1}{(x+1)^2}$$

Q) Find derivative of $y = \frac{\sin x}{\cos x}$

$$\frac{dy}{dx} = \frac{\cos x \frac{d}{dx}\sin x - \sin x \frac{d}{dx}\cos x}{\cos^2 x}$$

$$= \frac{\cos x(\cos x) - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x}.$$

* Chain Rule:-

Example:- $y = \log(\sin(\cos x^3))$

$$\frac{dy}{dx} = \frac{1}{\sin(\cos^2 x^3)} \times \cos(\cos x^3) \times (-\sin x^3) \times 3x^2$$

Ex2):- (i) $y = \log(\sin x)$

$$\frac{dy}{dx} = \frac{1}{\sin x} \times \frac{d}{dx}\sin x$$

$= \frac{\cos x}{\sin x}$

(ii) $y = \sin(\log x)$

$$\frac{dy}{dx} = \cos(\log x) \times \frac{d}{dx} \log x$$

$$= \frac{\cos(\log x)}{x}$$

(iii) $y = \sqrt{\sin x}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\sin x}} \times \frac{d}{dx} \sin x$$

$$= \frac{1}{2\sqrt{\sin x}} \times \cos x$$

$$= \frac{\cos x}{2\sqrt{\sin x}}$$

(iv) $y = \sqrt{(x^2+3x)}$

$$\frac{dy}{dx} = \cos(\log x) \times \frac{d}{dx} \log x$$

$$= \frac{\cos(\log x)}{x}$$

(v) $y = (ax+b)^5$

$x^5 \longrightarrow 5x^4$

$$\frac{dy}{dx} = 5(ax+b)^4 \times \frac{d}{dx}(ax+b)$$

$$= 5(ax+b)^4 \times (a+0)$$

$$= 5(ax+b)^4 a$$

1) Find $\frac{dy}{dx}$ for the following:-

(i) $y = 1 + 3x - x^2$

$\frac{dy}{dx} = 1 + 3x - 2x$

$= 1 + x$

(ii) $y = x^3 - 2x^2 - 3$

$\frac{dy}{dx} = 3x^2 - 2(2x) - 0$

$= 3x^2 - 4x$

(iii) $y = \frac{3}{x}$

$\frac{dy}{dx} = \frac{x(0) - 3(1)}{x^2}$

$= \frac{0 - 3x}{x^2}$

$= \frac{-3x}{x^2}$

(iv) $y = \sqrt{2x}\sqrt{2}\,x\,d\sqrt{x}$

$\frac{}{dx}$

$= \frac{\sqrt{2}d}{dx}$

$= \frac{\sqrt{2}x1}{2\sqrt{2}}\sqrt{2} \times \frac{1}{\sqrt{2}x\sqrt{2}x\sqrt{x}}$

$= \frac{1}{\sqrt{2}x} = \frac{1}{\sqrt{2}x}$

(v) $y = (ax+b)^2$

$= 2(ax+b)$

$= 2(ax+b) \times \frac{d}{dx}(ax+b)$

$= 2(ax+b) \times a + 0$

$= 2(ax+b)a.$

2) Evaluate y where $x=2$.

(i) $y = 8x - x^3$

$= 8(1) - 3x^2$

$= 8 - 3(2)^2$

$= 8 - 3 \times 4$

$= 8 - 12$

$= -4$

(ii) $y = ax^2 + bx + c$

$= a(2x) + b(1) + 0$

$= a(2 \times 2) + b$

$= 4a + b$.

---

* Rate of change with respect to time :-

dx/dt = rate of change of position with respect to time.

$\frac{dx}{dt}$ = Velocity = rate of change of position wrt.

$\frac{dv}{dt}$ = Rate of change of velocity wrt.

---

9) A person blows a bubble whose radius is increasing 0.1 cm every second. Find rate at which volume is increasing when $r = 1$ cm.

Sol) $\frac{dr}{dt} = \frac{0.1\,cm}{sec}$

$V = \frac{4}{3}\pi r^3$



$$\frac{dv}{dt} = \frac{d}{dt}\left(\frac{4}{3}\pi r^3\right)$$

$$= \frac{4\pi}{3}\left(\frac{dr^3}{dt}\right)$$

$$\frac{dv}{dt} = \frac{4}{3}\pi\, 3r^2 \frac{dr}{dt}$$

$$= \frac{4}{\cancel{3}}\pi \cancel{3} r^2 \frac{dr}{dt}$$

$$= 4\pi r^2 \frac{dr}{dt}$$

$$= 4\pi\, 1cm^2 \times 0.1 cm/sec$$

$$= 0.4\pi\, cm^3/sec$$

$$= 1.256\, cm^3/sec$$

Q) Water is poured into a cylindrical vessel with rate q. Find rate at which height of water level increases.



Sol) $Q = \frac{dv}{dt}$

$\frac{dh}{dt} = ?$

$V = \pi R^2 h$

$\frac{dv}{dt} = \frac{d}{dt}\pi R^2 h$

$q = \pi R^2 \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{q}{\pi R^2}$

* Maxima and Minima :-

At all points of maxima and minima, $\frac{dy}{dx} = 0$.

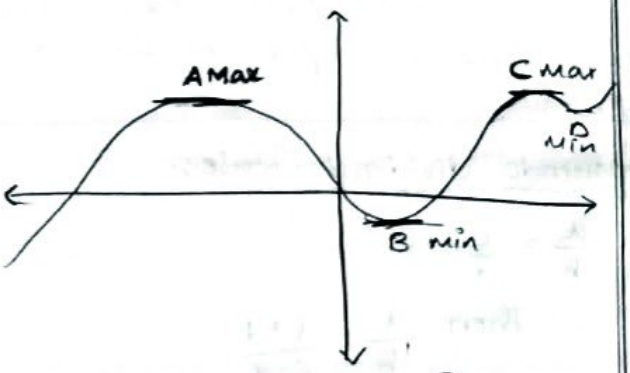
$y' = 0$

Slope decreases in maxima as x increases.
Slope increases in minima as x increases.

In maxima,

$\frac{d_{slope}}{dx} = -ve$

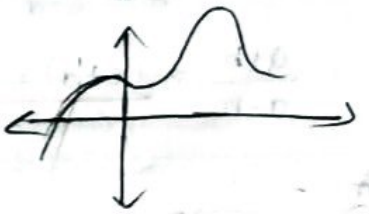
$\frac{d}{dx}$ slope.



In minima,

$\frac{d_{slope}}{dx} = +ve$

$\frac{d}{dx}$ slope.



Minima	Maxima
$\frac{d}{dx}\left(\frac{dy}{dx}\right)$	$\frac{d}{dx}\left(\frac{dy}{dx}\right)$
$\frac{d^2y}{dx^2} > 0$	$\frac{d^2y}{dx^2} < 0$
$y'' > 0$	$y'' < 0$.

Q) If $y = 5x^2 - 2x + 1$, find minimum value of y

Ans = 10.

Q) If $y = x^3 - 3x$, find points of minima & maxima.

For all points of max and min, $\frac{dy}{dx} = 0$.

$x = 1, x = -1$

for $x = 1$, $y'' = 6 > 0$

point of minima.

For $x = -1$, $y'' = 6x = -6 < 0$

point of maxima.

* Componendo dividendo Rule:-

If $\frac{a}{b} = \frac{c}{d}$,

Then $\frac{a}{b} = \frac{c+d}{c-d}$

$$\frac{a+b}{a-b} = \frac{(c+d)+(c-d)}{(c+d)-(c-d)} = \frac{c}{d}$$

* Logarithms:-

$b^x = y$

x → Power exponent

b → base

exponential form

Ex: $\log_{10} 1000 = 3$

$\log_{16}{}^{2} = \frac{1}{4}$

$\log_{4}{}^{4} = 1$

$\log_{2}{}^{1} = 0$

$\log_{5} 625 = 4$

$\log_{8}{}^{32} = \frac{5}{3}$

* Limitations :-

⇒ Base cannot be negative or equal to 1 ; or zero.

Base

- 10 (common base) ↓ If nothing is given at log, Ex : $\log_{(?)} 6$, we are to assume that it is 10. i.e, $\log_{10} 6$.
- e (Natural base) $e = 2.718\ldots$

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \ldots \infty$$

$5! = 5 \times 4 \times 3 \times 2 \times 1$.

$4! = 4 \times 3 \times 2 \times 1$.

* Rules of logarithms :-

1) $\log_b m + \log_b n = \log_b (mn)$ {This works only when bases are same.}

2) $\log_b m - \log_b n = \log_b \left(\frac{m}{n}\right)$ { " }

3) $\log_b a = \frac{\log_x a}{\log_x b}$ {where x can be anything except $x > 0$ and $x \neq 1$}

* Some Important Values of log :-

1) $\ln 2 \simeq 0.693$ (approximately = ≃)

2) $\log_{10} 2 \simeq 0.3010$

3) $\log 3 \simeq 0.4771$

4) $\ln 10 \simeq 2.303$

* $e^{2.303} = 10$ *

Q) Find the value of $\log 6000$

$= \log (2 \times 3 \times 1000)$

$= \log 2 + \log 3 + \log 1000$

$= 0.3010 + 0.4771 + 3$

$= 3.7781$

Q) Find $\log 50$

$= \log \left(\frac{100}{2}\right)$

$= \log 100 - \log 2$

$= 2 - 0.3010$

$= 1.6990$

$= 1.699 \approx 1.7$

Q) Take log to find derivative:-

(i) $y = a^x$

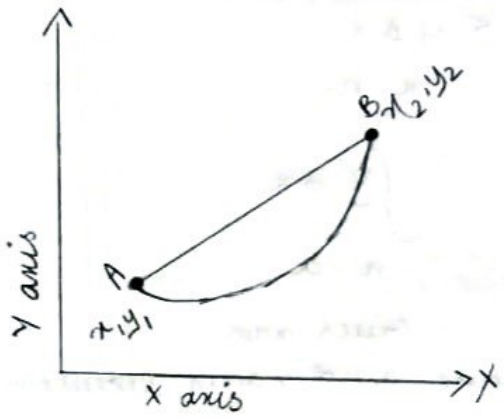
$\log y = \log a^x$

$\log x^p = p \log x$

* Average slope of a curve between two points :-

Average slope between points a and b $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{\Delta y}{\Delta x}$

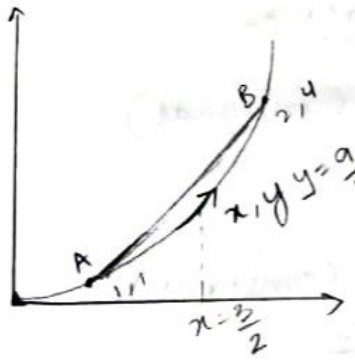


Q) Which point on a curve will have instantaneous slope = average slope ?

Find avg slope of curve $y = x^2$ b/w points $x = 1$ & $x = 2$

$A(1, 1)$, $B(2, 4)$

Avg. slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{2 - 1} = 3.$



* **Instantaneous slope** $\Rightarrow \frac{dy}{dx} = \frac{\Delta y}{\Delta x}$.

$\frac{dy}{dx}$ will give slope at a point, which is called instantaneous slope.

* Integration and area under curve :-

Area under the curve = Area of all rectangles

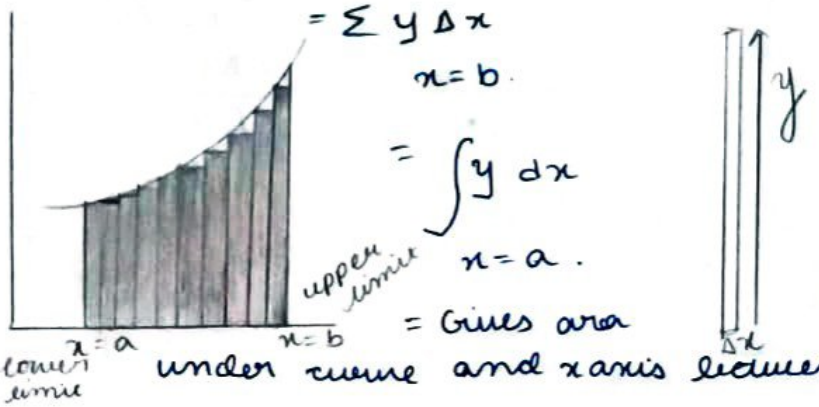
$$= \sum y \Delta x$$

$$= \int_{x=a}^{x=b} y \, dx$$

upper limit $x = b$

lower limit $x = a$

= Gives area under curve and x axis between the $x = a, x = b$.



$\int y dx$ → indefenite integral

$\int_a^b y dx$ → definite integral.

* Table of indefinite integrals :-

	y	$\int y \, dx$ (integral)
1,	1	x
2.	0	C (constant)
3,	x	$x^2/2$
4,	x^2	$x^3/3$
5.	x^3	$x^4/4$
6,	x^n	$x^{n+1}/n+1$

7,	$\frac{1}{x}$	$\ln x$
8,	e^x	e^x
9,	a^x	$a^x/\ln a$
10,	$\cos x$	$\sin x$
11,	$-\sin x$ $\sin x$	$\cos x$ $-\cos x$
12,	$\sec^2 x$	$\tan x$
13,	$\sec x \tan x$	$\sec x$
14,	$-\operatorname{cosec}^2 x$ $\operatorname{cosec}^2 x$	$\cot x$ $-\cot x$
15,	$-\cot x \csc x$ $\cot x \operatorname{cosec} x$	$\operatorname{cosec} x$ $\operatorname{cosec} x$

Q1) $\int x + x^2 + x^3 + 0 \, dx$

$= \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + C$

Q2) $\int \sqrt{x} + \frac{1}{\sqrt{x}} \, dx$

$= \int \left(x^{\frac{1}{2}} + x^{-\frac{1}{2}} \right) dx$

$= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C = \frac{x^{3/2}}{3/2} + \frac{x^{1/2}}{1/2} + C.$

g) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 dx$

$= \int (\sqrt{x})^2 + \left(\frac{1}{\sqrt{x}}\right)^2 + 2\sqrt{x}\frac{1}{\sqrt{x}}\, dx$

$= \int x + \frac{1}{x} + 2\, dx$

$= \frac{x^2}{2} + \ln x + 2x + C$

---

* Integration of (ax+b) type :-

g) $\int \cos(ax+b)\, dx$

$= \frac{\sin(ax+b)}{a} + C$

---

g) $\int \sin 2t + \cos 3t\, dt$

$= \frac{-\cos 2t}{2} + \frac{\sin 3t}{3} + C.$

---

g) $\int \sin^2 x\, dx$

$= \int \frac{1 - \cos 2x}{2}\, dx$

$= \int 1 - \cos 2x\, dx$

$$= \frac{1}{2}\left(\int 1-\cos 2x\, dx\right)$$

$$= \frac{1}{2} f\left(x - \frac{\sin 2x}{2} + C\right)$$

* Illustrations :-

Illustration - 13 :- pg-56 :-

(i) $4x^3 = \int 4x^3 dx = 4\left(\frac{x^{3+1}}{3+1}\right) + C = \frac{4x^4}{4} + C$

$= x^4 + C.$

(ii) $x - \frac{1}{x} = \int \left(x - \frac{1}{x}\right) dx = \int x\, dx - \int \frac{1}{x} dx$

$= \frac{x^2}{2} - \ln x + C$

(iii) $\frac{1}{2x+3} = \int \frac{dx}{2x+3} = \frac{\ln(2x+3)}{2} + C$

(iv) $\cos(4x+3) = \int \cos(4x+3)\, dx$

$= \frac{\sin(4x+3)}{4} + C$

(v) $\cos^2 x = \int \cos^2 x\, dx = \int \frac{2\cos^2 x}{2} dx$

$= \int \frac{(1+\cos 2x)}{2} dx = \frac{1}{2}\int dx + \frac{1}{2}\int \cos x\, dx$

$= \frac{x}{2} + \frac{1}{2}\frac{\sin 2x}{2} + C = \frac{x}{2} + \frac{1}{4}\sin 2x + C.$

* Beginner's Exercise-2 :- Pg-58 :-

Q1) (i) $\int 5 \, dx = 5x + C$

(ii) $\int (3x^2 - 5) \, dx$

$= \frac{3x^3}{3} - 5x + C$

$= x^3 - 5x + C$

(iii) $\int (x - 7) \, dx$

$= \int \frac{x^2}{2} - 7x$

$= \frac{x^2}{2} - 7x + C$

(iv) $\int (x + 1)(2 - x) \, dx$

$= \int x(2 - x) + 1(2 - x)$

$= 2x - x^2 + 2 - x$

$= \int x - x^2 + 2 \, dx$

$= \frac{x^2}{2} - \frac{x^3}{3} + 2x + C$

$= -\frac{x^3}{3} + \frac{x^2}{2} + 2x + C$

* Exercise-2- Pg-62:-

1) $\int \frac{dx}{\tan x + \cot x} = ?$

$= \frac{1}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$

$= \frac{1}{\left(\frac{\sin^2 x + \cos^2 x}{\sin x \cos x}\right)}$

$= \frac{\sin x \cos x}{1}$

$= \frac{\sin 2x}{2}$

$= \frac{-(\cos 2x^2)}{4}$

$= \frac{-\cos 2x^2}{4} + C$

2) $\int \frac{dx}{1-\sin x} = ?$

$= \int \frac{dx}{1-\sin x} \times \frac{1+\sin x}{1+\sin x} dx$

$= \int \frac{1+\sin x}{\cos^2 x} dx$

$= \int \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} dx$

$= \int \sec^2 x + \tan x \sec x dx$

$= \int \tan x + \sec x \, dx$

$= \int \tan x + \sec x dx + c.$

3) $\int \left(x + \frac{1}{x}\right)^3 dx$

$= \int (x)^3 + \left(\frac{1}{x}\right)^3 + 3(x)\left(\frac{1}{x}\right)\left(x + \frac{1}{x}\right)$

$= \int x^3 + \frac{1}{x^3} + 3x + \frac{3}{x}$

$= \int \frac{x^4}{4} + \frac{3x^2}{4} + \frac{3x^2}{2} + 3\log x$

$= \int \frac{x^4}{4} + \frac{3x^{-2}}{-2} + \frac{3x^2}{2} + 3\log x$

4) $\int \left(\cos \frac{x}{2} - \sin \frac{x}{2}\right)^2 dx$

$= \int \cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} - 2\sin \frac{x}{2} \cos \frac{x}{2} \, dx$

$= \int 1 - \sin x \, dx$

$= x + \cos x + C$

5) $\int \frac{5(x^6+1)}{x^2+1}\,dx$

$= \frac{5x^6+5}{x^2+1}$

$= \int x^5 - \frac{5x^3}{3} + 5x + C$

$= x^5 - \frac{5x^3}{3} + 5x + C.$

6) $\int \frac{x^2+x-6}{(x-2)(x-1)}\,dx$

$= x + 4\log(x-1) + C.$

7) $\int \frac{4t}{\sqrt{3t^2-7}}\,dt$

Let $\sqrt{3t^2-7} = x$

$3t^2 - 7 = x^2$

$d(3t^2-7) = dx^2$

$(3 \times 2t - 0)dt = 2x\,dx$

$6t\,dt = 2x\,dx$

$t\,dt = \frac{x\,dx}{3}$

$= \int \frac{4 \cancel{x} dx/3}{\cancel{x}}$

$= \frac{4}{3}\int dx$

$= \frac{4}{3}x + C$

$= \frac{4}{3}\sqrt{3t^2-7} + C$

10) $\int \frac{4y}{\sqrt{2y^2+1}}\,dy$

Let $\sqrt{2y^2+1} = x$

$2y^2+1 = x^2$

$d(2y^2+1) = dx^2$

$dy(2\times 2y + 0) = 2x\,dx$

$4y\,dy = 2x\,dx$

$y\,dy = \frac{x\,dx}{2}$

$= \int \frac{4 \times dx/2}{x}$

$= \int 4\,dx/2$

$= \frac{4}{2}\int dx$

$= \frac{4}{2}x + C$

$= \frac{4}{2}\sqrt{2y^2+1} + C.$

$= 2\sqrt{2y^2+1} + C$

---

9) $\int \frac{3}{(2-x)^2}\,dx$

Let $2-x = y$

$d(2-x) = dy$

$-dx = dy$

$\int \frac{3 \times (-dy)}{y^2} = -3\int y^{-2}\,dy$

$$= \frac{-3y^{-2+1}}{-2+1} + C$$

$$= \frac{+3y^{-1}}{+1} + C$$

$$= \frac{3y^{-1}}{1} + C$$

$$= \frac{3}{y} + C$$

$$= \frac{3}{2-x} + C$$

(16) $\int \frac{dx}{3x+2}$

$$= \int \frac{1}{ax+b}\, dx$$

$$= \frac{\ln(ax+b)}{a}\, dx$$

$$= \frac{\ln(3x+2)}{3} + C$$

$$= \frac{1}{3} \ln \frac{5}{2} = \ln \left(\frac{5}{3}\right)^{1/3}$$

$$= \ln \left(\frac{5}{3}\right)^{1/3}$$

Indefinite integrals	Definite integrals
$\int y\,dx = F(x)+C$	$\int_{x=a}^{x=b} y\,dx = [F(x)]_{x=a}^{x=b} - [F(x)]_{x=a}$ $= F(b) - F(a)$ Ex: $\int_1^2 x^2\,dx = \left[\frac{x^3}{3}\right]_{x=1}^{x=2}$ $= \frac{2}{3}^3 - \frac{1}{3}^3 = \frac{7}{3}$

Ex:- $\int_{30^\circ}^{90^\circ} \cos\theta\,d\theta$

$= [\sin\theta]_{30}^{90}$

$= \sin 90 - \sin 30^\circ$

$= 1 - \frac{1}{2}$

$= \frac{1}{2}.$

Pg-62-63- Questions continuation :-

8) $\int e^{(3x+10)}\,dx$

Let $3x+10 = y$

$d(3x+10) = dy$

$d(3x+0) = dy$

$3x\,dx = dy$

$x\,dx = \frac{dy}{3}$

$$= \int e^{dy/3}\, dx$$

$$= \int \frac{1}{3} e^{dy}\, dx$$

$$= \int \frac{1}{3} e^{(3x+10)} + C$$

13) $\int_{\pi}^{2\pi} \theta\, d\theta$

$$= \int_{\pi}^{2\pi} \frac{\theta^2}{2}\, d\theta$$

$$= \left[\frac{\theta^2}{2}\right]_{\pi}^{2}$$

$$= \frac{(2\pi)^2}{2} - \frac{\pi^2}{2}$$

$$= \frac{4\pi^2}{2} - \frac{\pi^2}{2} = \frac{3\pi^2}{2}$$

14) $\int_{0}^{\pi} \cos x\, dx$

$$= [\sin x]_{0}^{\pi}$$

$$= \sin\pi - \sin 0$$

$$= \sin\pi - 0$$

$$= \sin\pi,$$

where π is some constant

$$= \sin 0$$

$$= 0.$$

15) $\int_0^{\sqrt{\pi}} x \sin x^2 dx$

$= \int_0^{\sqrt{\pi}} x \sin x^2 dx$

$= \int_0^{\sqrt{\pi}} \sin x^2 dx^2 \quad x^2 = y$

$= \frac{1}{2} \int \sin y \, dy \quad dx^2 = dy$

$= \left[\frac{-\cos x^2}{2} \right]_0^{\sqrt{\pi}}$

$= \frac{-\cos x^2 \sqrt{\pi}}{2} + \frac{\cos x^2 0}{2}$

$= \frac{-\cos x^2 \sqrt{\pi} + 0}{2}$

$= \frac{-\cos x^2 \sqrt{\pi}}{2}$

where π is some constant

$= \frac{-\cos x^2 \times 0}{2}$

$= 1$

17) $y = 3x^2$

$= \int y \, dx$

$= \int_0^b 3x^2 dx$

$= 3 \left[\frac{x^{2+1}}{2+1} \right]_0^b$

$= (x^3)_0^b = b^3 - 0^3 = b^3$

18) $y = 2x$

$$= \int y\,dx$$

$$= \int_0^b 2x\,dx$$

$$= 2\left(\frac{x^2}{2}\right)_0^b$$

$$= (x^2)_0^b$$

$$= b^2 - 0^2$$

$$= b^2.$$

---

19) $y = \frac{x}{2} + 1$

$$= \int y\,dx$$

$$= \int_0^b \frac{x}{2} + 1\,dx$$

$$= \int_0^b \frac{x+2}{2}\,dx$$

$$= \int_0^b \frac{x^2}{4} + x$$

$$= \frac{b^2}{4} + b - 0$$

$$= \frac{b^2 + 4b}{4}$$

$$= \frac{b(b+4)}{4}$$

20) Use a definite integral to find the area of the region between the given curve and the axis(x) on the interval $[0, \pi]$: $y = \sin x$.

Sol) $y = \sin x$

$$= \int_{\pi}^{\circ} y \, dx$$

$$= \int_{\pi}^{\circ} \sin x \, dx$$

$$= \int_{\pi}^{\circ} -\cos x \, dx$$

$$= \int_{\pi}^{\circ} (-\cos x)(dx)$$

$$= \int (-\cos x)^{\circ} - (-\cos x)^{\pi}$$

$$= \int 1 + \cos x^{\pi}$$

where π is some constant.

$$= \int 1 + \cos x^{\circ}$$

$$= \int 1 + 1$$

$$= 2.$$

11) $\int (2x^2 + e^x) dx$

$$= \frac{2x^3}{3} + e^x + C.$$

* Different units of length :-

⇒ For small units :-

→ 1 Fermi = 10^{-15} m
(1 femto meter)

→ Size of nuclei is measured in fermi

→ 1 Angstrom = 10^{-10} m. (size of atoms & molecules)
$1\,A^{\circ} = 10^{-10}$ m

→ 1 nanometer = 10^{-9} m
1 nm = $10\,A^{\circ}$

→ 1 micron = 10^{-6} m
= 1 Mm
(micro meter)

⇒ For Very Large scale :-

⇒ 1 lightyear = $3 \times 10^{8} \frac{m}{s} \times 365 \times 24 \times 3600$ Sec

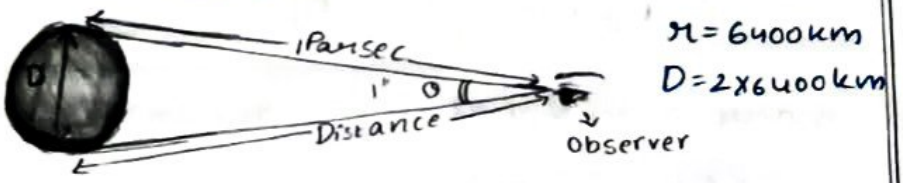
$= 9.46 \times 10^{15}$ m/s

$= 9.46 \times 10^{12}$ km

→ 1 Astronomical unit (AU) = Avg. dist. b/w sun and Earth.
≈ 150 million km
≈ 150×10^{6} km
(149.6×10^{6} km)

1 million = 10^{6}.
1 billion = 10^{9}
1 trillion = 10^{12}

→ 1 Parsec is the distance at which the Earth will subtend an angle of 1 arc second on the Observer.



$$\theta \approx \frac{\text{Diameter}}{\text{Distance}} = \frac{2 \times 6400\ km}{\text{distance}}$$

$$1 \text{ arc sec} = \left(\frac{1}{3600} \times \frac{\pi}{180}\right) \text{rad}$$

$$(1 \text{ parsec})(\text{distance}) = \frac{\text{Diameter}}{\theta \text{ in rad}}$$

$$= \frac{2 \times 6400\ km}{\frac{1}{3600} \times \frac{\pi}{180}}$$

$$= 3.08 \times 10^{13}\ km$$

$$= 30.8 \times 10^{12}\ km$$

$$= 30.8 \text{ trillion km}$$

8) How many light years are in 1 parsec?

$$1 \text{ parsec} = _____ \text{ ly}$$

$$\frac{1 \text{ parsec}}{\text{ly}} = _____$$

$$= \frac{30.8 \text{ Trillion km}}{9.46 \text{ Trillion km}}$$

$= 3.26$

$\therefore$ 1 parsec $= 3.26$ ly.

9) How many astronomical units are in 1 ly?

1 ly = ____ AU

$\frac{1\ ly}{AU} =$ ____

$AU = \frac{9.46 \times 10^{12}\ km}{150 \times 10^{6}\ km}$

$AU = \frac{9.46 \times 10^{6}\ km}{150}$

$AU = 63.241 \times 10^{4}\ km$

$AU = 63241\ AU$

1 ly = 63241 AU.

$\Rightarrow$ 1 Parsec > 1 ly > 1 AU

* Different units of masses:-

-> Pico (p) = 10^{-12}
-> Nano (n) = 10^{-9}
-> Micro (mu) = 10^{-6}
-> milli (m) = 10^{-3}
-> deci (d) = 10^{-1}

} Small prefixes.

→ Deca = 10
→ kilo = 10^3 = 1000
→ Mega = 10^6
→ Giga = 10^9

} Bigger prefixes.

⇒ 1 Quintal = 100kg
⇒ 1 Tonne = 1000kg

* Accuracy and Prescision :-

⇒ Accuracy - Degree of closeness of measured value to the actual value

⇒ Prescision - Dependent on how close the measurements are to eachother. It also depends on resolution of the instrument.

Example :



P	Q	R
71 kg	70 kg	68.25 kg
70 kg	65 kg	68.35 kg
71 kg	68 kg	68.77 kg
72 kg	75 kg	68.74 kg
71 kg	69.5kg	68.53 kg
Most accurate		Most precise

* Types of Errors :-

Systematic Error	Random Error
* En: Error in procedure	* Random errors are due to unknown reasons
* Cause can be found	* Unknown cause
* Error can be in the instrument	* They tend to be +ve and -ve with equal tendencies.
* Manual / Personal error	* Equal tendency to be higher than true value
* They favour tendency in one direction	* Equal tendency to be lower than true value
* zero error = Actual value = Measured value − Zero error.	* Take multiple measurements and find the average
* En: Zero error of a weighing m/c is −50g. What is actual reading if measured value is 375g? = 375 g − (−50g) = 425 g.	En: Least count error A: 00·0, 00·1 → 12·0 ± 0·1 B: 00·00, 00·01 → 12·00 ± 0·01 C: 00·000, 00·001 → 12·000 ± 0·001 sec

* Types of errors based on definition :-

1, Absolute error :- Measured value − True value

ΔA ↓ $A+\Delta A$ ↓ A ↓

2, Relative error :- $\frac{\text{Absolute error}}{\text{True value}} = \frac{\Delta A}{A} = \frac{\Delta A}{A}$ (No units / No dimension)

3, % error $= \frac{\Delta A}{A} \times 100$

* Propogation of error :-

→ Sum / Difference rule :- $g = A+B-C-D$

$$\Delta g = \Delta A + \Delta B + \Delta C + \Delta D$$

→ Product / Division rule :- $g = \frac{AB^2}{C^3D^4}$

$$\frac{\Delta g}{g} = \frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta B}{B} + \frac{\Delta C}{C} + \frac{\Delta C}{C} + \frac{\Delta C}{C} + \frac{\Delta D}{D} + \frac{\Delta D}{D} + \frac{\Delta D}{D} + \frac{\Delta D}{D}$$

$$\Rightarrow \frac{\Delta Q}{Q} = \frac{\Delta A}{A} + \frac{2\Delta B}{B} + \frac{3\Delta C}{C} + \frac{4\Delta D}{D}$$

$$\% g = \% A + 2\% B + 3\% C + 4\% D.$$